

Step 1a Cache Builder Failure Analysis

Data source: data/deviation_experiment/cache_eval_results.json.

Scope: only the three cache-builder configs are included:

- clip_w7_d4
- max_pool_w3_d5
- spatial16_w8_d4

All inference_* rows are intentionally excluded. Each builder has 500 rows: 10 libero_spatial tasks x 50 default .pruned_init states. A unit below is identified by (task_id, init_state_idx).

Compared Configs

These are the exact non-inference cache-builder configs compared in this analysis. trajectory_depth is the trajectory length used by weighted_rrf_knn; trajectory_weights are the per-depth weights.

config	key builder type	key weights	trajectory depth	trajectory weights
clip_w7_d4	clip	vision_0=0.1, vision_1=0.1, prompt_emb=0.0, robot_state=0.8	4	[0.4, 0.3, 0.2, 0.1]
max_pool_w3_d5	cp1_max_pool	vision_0=0.5, vision_1=0.0, prompt_emb=0.0, robot_state=0.5	5	[0.35, 0.25, 0.2, 0.12, 0.08]
spatial16_w8_d4	cp1_spatial_pool_16	vision_0=0.5, vision_1=0.25, prompt_emb=0.0, robot_state=0.25	4	[0.4, 0.3, 0.2, 0.1]

All three use weighted_rrf_knn, top_k=1, step_filter=all, rrf_k=60, and the same robot-state similarity transform: $l2 \rightarrow \exp(\tau=0.334717)$.

Summary

Overall success rates are close, but the failures are not interchangeable:

builder	success	failure
clip_w7_d4	337/500 (67.4%)	163
max_pool_w3_d5	348/500 (69.6%)	152
spatial16_w8_d4	346/500 (69.2%)	154

Across the 500 shared init units:

category	units
all 3 pass	223
exactly 1 builder fails	136
exactly 2 builders fail	90
all 3 builders fail	51

Interpretation:

- The 51 all-fail units are likely genuinely hard init states for this policy/cache setting.
- The 90 exact-two failures show builder-specific complementarity: one builder can recover a nontrivial number of cases where the other two fail.
- The 136 exact-one failures are builder-specific weaknesses rather than task-inherent failures.

Core conclusion:

The interesting result is that similar aggregate success rates do not imply equivalent keybuilders. The three builders land around the same total success rate, but they do so by succeeding and failing on different task/init slices. This suggests two things at once:

- Some tasks or init states are broadly hard for this cache-builder family:

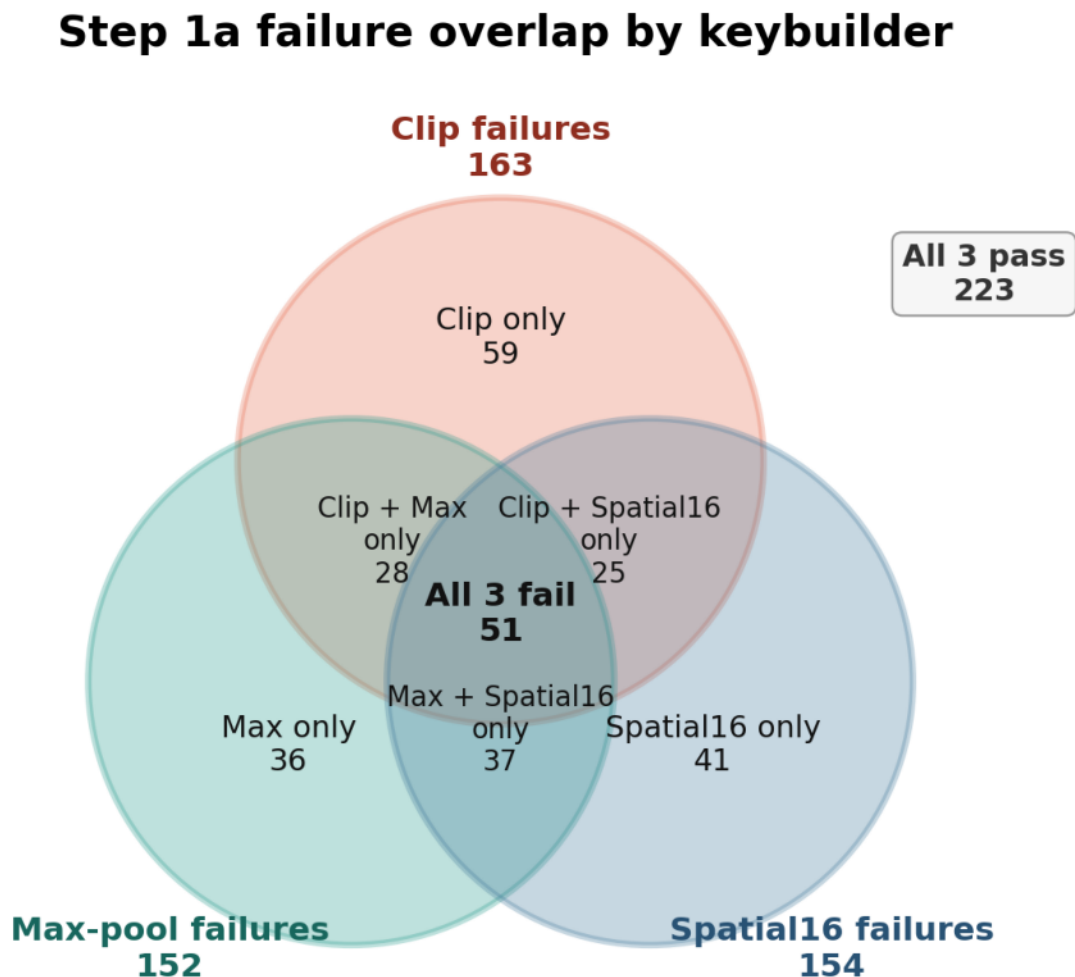
51 init units fail under all three builders.

- Other failures are builder-specific: 136 init units fail under exactly one

builder, and 90 fail under exactly two builders. This is evidence that the keybuilders have different task/init preferences rather than simply being noisy variants of the same behavior.

The clearest examples are task 1 and task 9. On task 1, clip_w7_d4 reaches 76% while max_pool_w3_d5 drops to 50%. On task 9, the pattern flips: max_pool_w3_d5 reaches 72%, spatial16_w8_d4 reaches 60%, and clip_w7_d4 drops to 32%. So the builders have similar aggregate performance, but they are not interchangeable at the task level.

Failure-set Venn diagram:



Universe: 500 shared (task_id, init_state_idx) units. Circles are failures, not successes.

Step 1a failure-set Venn diagram

The diagram is generated by plot_step1a_failure_venn.py. Each circle is a failure set, not a success set. The exact region counts are: C only=59, M only=36, S only=41, C+M only=28, C+S only=25, M+S only=37, and C+M+S=51.

Failure Overlap

Pairwise intersections including the all-3 failures:

pair	shared failures
clip_w7_d4 + max_pool_w3_d5	79
clip_w7_d4 + spatial16_w8_d4	76
max_pool_w3_d5 + spatial16_w8_d4	88

Exact overlap after excluding all-3 failures:

exact category	units
clip_w7_d4 + max_pool_w3_d5 only	28
clip_w7_d4 + spatial16_w8_d4 only	25
max_pool_w3_d5 + spatial16_w8_d4 only	37

Unique builder-only failures:

builder-only failure	units
clip_w7_d4 only	59
max_pool_w3_d5 only	36
spatial16_w8_d4 only	41

clip_w7_d4 has the most unique failures, mostly because task 9 contributes 16 clip-only failures. max_pool_w3_d5 has the fewest unique failures, which is consistent with its slightly higher overall success rate.

Task-Level Success

task	clip_w7_d4	max_pool_w3_d5	spatial16_w8_d4	best	spread
0	37/50 (74%)	41/50 (82%)	35/50 (70%)	max_pool_w3_d5	6
1	38/50 (76%)	25/50 (50%)	31/50 (62%)	clip_w7_d4	13
2	36/50 (72%)	41/50 (82%)	40/50 (80%)	max_pool_w3_d5	5
3	40/50 (80%)	39/50 (78%)	38/50 (76%)	clip_w7_d4	2
4	32/50 (64%)	27/50 (54%)	31/50 (62%)	clip_w7_d4	5
5	33/50 (66%)	33/50 (66%)	37/50 (74%)	spatial16_w8_d4	4
6	37/50 (74%)	36/50 (72%)	39/50 (78%)	spatial16_w8_d4	3
7	32/50 (64%)	34/50 (68%)	34/50 (68%)	max_pool_w3_d5 / spatial16_w8_d4	2
8	36/50 (72%)	36/50 (72%)	31/50 (62%)	clip_w7_d4 / max_pool_w3_d5	5
9	16/50 (32%)	36/50 (72%)	30/50 (60%)	max_pool_w3_d5	20

Main task-level takeaways:

- clip_w7_d4 is best on tasks 1, 3, and 4, and tied for best on task 8.
- max_pool_w3_d5 is best on tasks 0, 2, and 9, tied on tasks 7 and 8, and is the most stable overall.
- spatial16_w8_d4 is best on tasks 5 and 6, tied on task 7.
- Task 9 is the clearest builder-specialization case: max_pool_w3_d5 gets 72%, spatial16_w8_d4 gets 60%, while clip_w7_d4 drops to 32%.
- Task 1 is the opposite pattern: clip_w7_d4 gets 76%, while max_pool_w3_d5 drops to 50%.

Task-Level Failure Categories

C = clip_w7_d4, M = max_pool_w3_d5, S = spatial16_w8_d4. The pair columns are exact-two failures, so they exclude the all-3 failures.

task	all pass	any fail	C only	M only	S only	C+M only	C+S only	M+S only	all 3 fail
0	27	23	5	2	8	1	1	0	6
1	17	33	1	7	5	6	2	9	3
2	26	24	6	4	6	4	3	0	1
3	32	18	3	3	2	0	2	3	5
4	16	34	5	7	2	3	4	7	6
5	23	27	5	5	2	4	3	3	5
6	24	26	8	6	3	1	1	4	3
7	26	24	5	0	2	3	1	4	9
8	21	29	5	2	8	3	2	5	4
9	11	39	16	0	3	3	6	2	9

Task-level interpretation:

- Task 9 has the most any-fail units (39/50), the most all-3 failures tied with task 7 (9), and the most clip-only failures (16). This is the strongest evidence that clip_w7_d4 is specifically weak on this task.
- Task 4 has high any-fail count (34/50) and a broad spread across exact pairs and unique failures. This looks generally difficult rather than one-builder-specific.
- Task 1 has many M+S only failures (9) and M only failures (7), while clip_w7_d4 has only 1 unique failure. This suggests the CLIP-based builder handles this task's init variation better.
- Task 7 has 9 all-3 failures but zero max-only failures. Its hard cases are shared, and max_pool_w3_d5 does not introduce much isolated damage there.
- Task 3 is the easiest/most consistent task: only 18/50 units fail at least one builder, and the best-worst spread is just 2 successes.

All-3 Failure Inits

These 51 units failed under all three builders:

- T0: [2, 13, 18, 24, 27, 39]
- T1: [14, 35, 36]
- T2: [45]
- T3: [2, 14, 24, 38, 42]
- T4: [20, 27, 35, 42, 45, 47]
- T5: [11, 23, 28, 31, 45]
- T6: [9, 24, 34]
- T7: [10, 11, 26, 27, 41, 42, 45, 46, 49]
- T8: [4, 12, 29, 33]
- T9: [6, 9, 10, 18, 29, 37, 41, 46, 49]

Exact-Two Failure Inits

These units failed under exactly two builders and succeeded under the third.

clip_w7_d4 + max_pool_w3_d5 only

28 units:

- T0: [1]
- T1: [2, 17, 30, 33, 37, 43]
- T2: [19, 22, 33, 48]
- T4: [18, 30, 39]
- T5: [3, 15, 24, 49]
- T6: [7]
- T7: [35, 38, 40]
- T8: [34, 39, 43]
- T9: [42, 43, 45]

clip_w7_d4 + spatial16_w8_d4 only

25 units:

- T0: [11]
- T1: [20, 24]
- T2: [2, 12, 36]
- T3: [7, 9]
- T4: [5, 13, 25, 26]
- T5: [9, 33, 35]
- T6: [26]
- T7: [15]
- T8: [13, 26]
- T9: [0, 2, 5, 13, 22, 48]

max_pool_w3_d5 + spatial16_w8_d4 only

37 units:

- T1: [1, 4, 12, 16, 22, 32, 38, 42, 48]
- T3: [18, 27, 29]
- T4: [1, 3, 8, 10, 15, 17, 23]
- T5: [1, 4, 18]
- T6: [2, 6, 36, 49]
- T7: [13, 31, 37, 39]
- T8: [1, 2, 32, 44, 49]
- T9: [3, 35]

Unique Failure Inits

These units failed under exactly one builder and succeeded under the other two.

clip_w7_d4 only

59 units:

- T0: [5, 12, 15, 17, 26]
- T1: [31]
- T2: [0, 9, 25, 26, 34, 47]
- T3: [19, 20, 28]

- T4: [7, 12, 16, 24, 48]
- T5: [6, 19, 21, 22, 32]
- T6: [3, 10, 21, 22, 27, 35, 44, 48]
- T7: [2, 33, 43, 44, 47]
- T8: [5, 10, 18, 35, 46]
- T9: [1, 4, 8, 11, 12, 14, 15, 17, 19, 25, 28, 32, 36, 39, 44, 47]

max_pool_w3_d5 only

36 units:

- T0: [8, 45]
- T1: [10, 13, 29, 39, 44, 45, 47]
- T2: [5, 14, 20, 41]
- T3: [5, 17, 48]
- T4: [11, 22, 29, 37, 38, 40, 46]
- T5: [0, 12, 17, 30, 36]
- T6: [1, 4, 8, 25, 28, 31]
- T8: [21, 25]

spatial16_w8_d4 only

41 units:

- T0: [4, 7, 14, 21, 22, 23, 46, 47]
- T1: [3, 34, 41, 46, 49]
- T2: [4, 8, 11, 24, 28, 35]
- T3: [35, 47]
- T4: [0, 33]
- T5: [41, 44]
- T6: [14, 16, 17]
- T7: [24, 36]
- T8: [0, 3, 6, 9, 16, 23, 37, 42]
- T9: [23, 33, 34]

Task Map

task	description
0	pick up the black bowl between the plate and the ramekin and place it on the plate
1	pick up the black bowl next to the ramekin and place it on the plate
2	pick up the black bowl from table center and place it on the plate
3	pick up the black bowl on the cookie box and place it on the plate
4	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate
5	pick up the black bowl on the ramekin and place it on the plate
6	pick up the black bowl next to the cookie box and place it on the plate
7	pick up the black bowl on the stove and place it on the plate
8	pick up the black bowl next to the plate and place it on the plate
9	pick up the black bowl on the wooden cabinet and place it on the plate